



Subject : Physics

Standard : 11

Total Mark : 200

Kinematics

Paper Set : 1

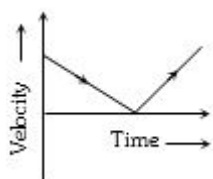
Date : 24-08-2026

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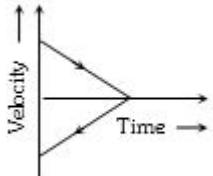
Physics - Section A (MCQ)

- (1) Which of the following represents the velocity-time graph for a ball thrown vertically upwards (neglecting air resistance)?

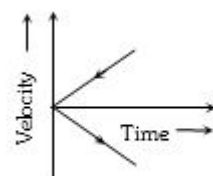
(A)



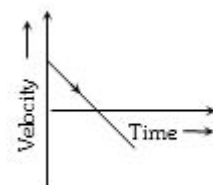
(B)



(C)



(D)



- (2) If the kinetic energy of a second ball at its maximum height is K , what will be the kinetic energy of the first ball at its maximum height? (Assuming the first ball is thrown vertically upwards).

(A) $4K$ (B) $3K$
(C) $2K$ (D) 0

- (3) The horizontal range is four times the maximum height attained by a projectile. The angle of projection is °

(A) 90 (B) 60
(C) 45 (D) 30

- (4) If the velocity v of a particle moving along a straight line decreases linearly with its displacement s from 20 m/s to a value approaching zero at $s = 30 \text{ m}$, then the acceleration of the particle at $s = 15 \text{ m}$ is

(A) $2/3 \text{ m/s}^2$ (B) $-2/3 \text{ m/s}^2$
(C) $20/3 \text{ m/s}^2$ (D) $-20/3 \text{ m/s}^2$

- (5) Statement (I): An object subjected to velocities $\vec{v}_1$ and $\vec{v}_2$ has a resultant velocity with magnitude $|\vec{v}| = |\vec{v}_1| + |\vec{v}_2|$.
Statement (II): The magnitude of displacement is either less than or equal to the path length of an object between two points.

Statement (III): The instantaneous acceleration is the limiting value of the average acceleration as the time interval approaches zero.

Which of the following is correct?

[TS EAMCET 2022]

(A) Statements I, II, III are true
(B) Statements I, II are true, but Statement III is false
(C) Statements II, III are true, but Statement I is false
(D) Statements I, II, III are false

- (6) A projectile is thrown with velocity u at an angle of 15° and has a range R . What will be the range of the projectile if it is thrown with velocity $2u$ at an angle of 45° (in R)?

(A) 12 (B) 3
(C) 8 (D) 4

- (7) A person can throw a ball up to a maximum range of 100 m . How high above the ground can he throw the same ball (in m)?

[JEE MAIN 2022]

(A) 25 (B) 50
(C) 100 (D) 200

- (8) A projectile object is thrown in the upward direction making an angle of 60° with the horizontal with velocity of 140 m/s . Then, the time after which its velocity makes an angle 30° with the horizontal is (Take, $g = 10 \text{ m/s}^2$)

[TS EAMCET 2021]

(A) $\frac{14}{\sqrt{3}} \text{ s}$ (B) $7\sqrt{3} \text{ s}$
(C) $14\sqrt{3} \text{ s}$ (D) $\frac{7}{\sqrt{3}} \text{ s}$

- (9) The ratio of minimum kinetic energies of two projectiles of the same mass is 4 : 1. The ratio of the maximum heights attained by them is also 4 : 1. The ratio of their ranges would be: (in : 1)

(A) 16 (B) 4
(C) 8 (D) 2

- (10) The maximum horizontal range of a projectile is 400 m . What is the maximum height attained by it (in m)?

(A) 100 (B) 200
(C) 400 (D) 800

- (11) What is the path followed by a moving body, on which a constant force acts in a direction other than the initial velocity (i.e., excluding parallel and antiparallel directions)?

- (A) Straight line (B) Parabolic
(C) Circular (D) Elliptical
- (12) A particle moves along the x -axis as $x = 4(t - 2) + a(t - 2)^2$. Which of the following statements is true?
(A) The initial velocity of the particle is 4.
(B) The acceleration of the particle is $2a$.
(C) The particle is at the origin at $t = 0$.
(D) None of the above.
- (13) Two paper screens A and B are separated by a distance of 100 m . A bullet penetrates A and B at points P and Q respectively, where Q is 10 cm below P . If the bullet is travelling horizontally at the time of hitting A , the velocity of the bullet at A is nearly m/s .
(A) 100 (B) 200
(C) 600 (D) 700
- (14) A particle is projected horizontally from a tower with velocity 10 m/s . Taking $g = 10\text{ m/s}^2$, match the following two columns at time $t = 1\text{ s}$.

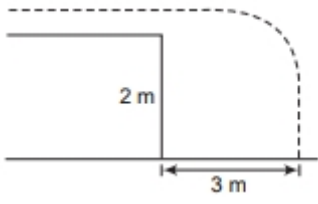
Column I	Column II
(A) Horizontal component of velocity	(p) 5 SI unit
(B) Vertical component of velocity	(q) 10 SI unit
(C) Horizontal displacement	(r) 15 SI unit
(D) Vertical displacement	(s) 20 SI unit

- (A) ($A \rightarrow q, B \rightarrow q, C \rightarrow q, D \rightarrow p$) (B) ($A \rightarrow q, B \rightarrow r, C \rightarrow q, D \rightarrow p$)
(C) ($A \rightarrow q, B \rightarrow s, C \rightarrow q, D \rightarrow p$) (D) ($A \rightarrow s, B \rightarrow q, C \rightarrow q, D \rightarrow p$)
- (15) When a ball is thrown with a velocity of 50 ms^{-1} at an angle 30° with the horizontal, it remains in the air for how many seconds? (Take $g = 10\text{ ms}^{-2}$) [AP EAMCET 2021]
(A) 5 (B) 2.5
(C) 1.25 (D) 0.625
- (16) The position of an object moving along the x -axis is given by $x = a + bt^2$, where $a = 8.5\text{ m}$, $b = 2.5\text{ ms}^{-2}$, and t is measured in seconds. What is the average velocity between $t = 2.0\text{ s}$ and $t = 4.0\text{ s}$ (in ms^{-1})?
(A) 20 (B) 10
(C) 25 (D) 15
- (17) A particle moves in the east direction with 15 m/s for 2 s , then moves northward with 5 m/s for 8 s . The average velocity of the particle is:
(A) 5 m/s due $E - 37^\circ - N$
(B) 5 m/s due $N - 37^\circ - E$
(C) 7 m/s due $S - 37^\circ - W$
(D) 10 m/s due $N - 37^\circ - E$
- (18) What is the angle between the velocity and acceleration of a projectile at its maximum height (in $^\circ$)?
(A) 0 (B) 45
(C) 90 (D) 180

- (19) If a stone is released from a balloon rising with acceleration a at the instant when its velocity is v , then immediately after release, the acceleration and velocity of the stone are
(A) a (upward), v (upward)
(B) g (upward), v (upward)
(C) g (downward), v (upward)
(D) $(g - a)$ (downward), v (upward)
- (20) A ball is projected upwards from a height h above the surface of the earth with velocity v . The time at which the ball strikes the ground is [AP EAMCET 2020]
(A) $\frac{v}{g} \left[1 + \sqrt{1 + \frac{2gh}{v^2}} \right]$ (B) $\frac{v}{g} \left[1 - \sqrt{1 + \frac{2gh}{v^2}} \right]$
(C) $\frac{v}{g} \left[1 + \sqrt{1 + \frac{2gh}{v^2}} \right]$ (D) $\frac{v}{g} \left[1 + \sqrt{v^2 + \frac{2g}{v^2}} \right]$
- (21) A particle covers half of its total distance with speed v_1 and the rest half distance with speed v_2 . Its average speed during the complete journey is [AIPMT 2011]
(A) $\frac{v_1 + v_2}{2}$ (B) $\frac{v_1 v_2}{v_1 + v_2}$
(C) $\frac{2v_1 v_2}{v_1 + v_2}$ (D) $\frac{v_1 + v_2}{3}$
- (22) A particle of mass 1 kg is moving along the x -axis, where its velocity depends on the x -coordinate as $v = 2\sqrt{x}\text{ m/s}$. Find the net force acting on the particle 2 s after the start of motion.
(A) 1 (B) 2
(C) 4 (D) 1.5
- (23) A body is falling freely from a point A at a certain height from the ground and passes through points B, C , and D (vertically as shown below) such that $BC = CD$. The time taken by the particle to move from B to C is 2 s and from C to D is 1 s . The time taken to move from A to B in seconds is [TS EAMCET 2016]



- (A) 0.6 (B) 0.5
(C) 0.2 (D) 0.4
- (24) A train has a speed of 60 km/h for the first one hour and 40 km/h for the next half hour. Its average speed in km/h is
(A) 50 (B) 53.33
(C) 48 (D) 70
- (25) When the velocity of a body is variable, then:

- (A) Its speed may be constant
(B) Its acceleration may be constant
(C) Its average acceleration may be constant
(D) All of these
- (26) A piece of marble is projected from the Earth's surface with a velocity of $19.6\sqrt{2} \text{ m/s}$ at an angle of 45° . 2 s later, its velocity makes an angle α with the horizontal, where α is (in $^\circ$)
(A) 45 (B) 30
(C) 60 (D) 0
- (27) The ratio of times taken by a freely falling body to travel the first 5 m, second 5 m, and third 5 m distances is:
[TS EAMCET 2025]
(A) $1 : \sqrt{2} : \sqrt{3}$ (B) $1 : \sqrt{2} - 1 : \sqrt{3} - \sqrt{2}$
(C) $1 : \sqrt{3} : \sqrt{5}$ (D) $1 : 2 : 3$
- (28) An object is projected such that it attains maximum range. Another body is projected to reach maximum height. If both objects reach the same maximum height, what is the ratio of their initial velocities?
[AP EAMCET 2024]
(A) 2 : 1 (B) $\sqrt{2} : 1$
(C) $1 : \sqrt{2}$ (D) 1 : 2
- (29) A ball is dropped downwards. After 1 second, another ball is dropped downwards from the same point. What is the distance between them after 3 seconds?
(A) 25 (B) 20
(C) 50 (D) 9.8
- (30) An aeroplane moving horizontally with a speed of 720 km/h drops a food packet while flying at a height of 396.9 m . The time taken by the food packet to reach the ground and its horizontal range are (Take $g = 9.8 \text{ m/s}^2$):
(A) 3 s and 2000 m (B) 5 s and 500 m
(C) 8 s and 1500 m (D) 9 s and 1800 m
- (31) Water is flowing from a horizontal pipe fixed at a height of 2 m from the ground. If it falls at a horizontal distance of 3 m as shown in the figure, the speed of water when it leaves the pipe is ms^{-1} (take $g = 9.8 \text{ ms}^{-2}$).
- 
- (A) 2.4 (B) 4.7
(C) 7.4 (D) 6.2
- (32) The angle of projection of a projectile for which its initial kinetic energy becomes half at its maximum height is (in $^\circ$)
[AP EAMCET 2017]
(A) 90 (B) 60
(C) 45 (D) 30
- (33) The time of flight of an object projected with speed 20 m/s at an angle 30° with the horizontal is s.
(A) 1 (B) 4
(C) 2 (D) 6
- (34) A particle initially at rest is moving along a straight line with an acceleration of 2 m/s^2 . At a time of 3 s after the beginning of motion, the direction of acceleration is

reversed. The time from the beginning of the motion in which the particle returns to its initial position is
[TS EAMCET 2025]

- (A) $(3 + \sqrt{3}) \text{ s}$ (B) $(2 + \sqrt{2}) \text{ s}$
(C) $3(1 + \sqrt{2}) \text{ s}$ (D) $2(3 + \sqrt{3}) \text{ s}$
- (35) A boy walks to his school at a distance of 6 km with a constant speed of 2.5 km/h and walks back with a constant speed of 4 km/h . His average speed for the round trip, expressed in km/h , is:
(A) 24/13 (B) 40/13
(C) 3 (D) 1/2
- (36) With what velocity should a ball be projected vertically so that the distance covered by it in the 5th second is twice the distance it covers in its 6th second? (Take $g = 10 \text{ m/s}^2$)
(A) 58.8 (B) 49
(C) 65 (D) 19.6
- (37) A ball of mass 0.2 kg is thrown from a height of 1 m with an initial velocity of $\sqrt{10} \text{ m/s}$ at an angle of 45° with the horizontal. Assuming acceleration due to gravity $g = 10 \text{ m/s}^2$, the modulus of the momentum increment during the total time of motion in $\text{kg} \cdot \text{m/s}$ is:
[TS EAMCET 2018]
(A) $\frac{2+\sqrt{10}}{\sqrt{10}}$ (B) $\frac{1+\sqrt{10}}{\sqrt{5}}$
(C) $\frac{1+\sqrt{5}}{\sqrt{5}}$ (D) $\frac{\sqrt{5}-1}{\sqrt{5}}$
- (38) A particle moving along the X-axis has acceleration f at time t given by $f = f_0 (1 - \frac{t}{T})$, where f_0 and T are constants. The particle at $t = 0$ has zero velocity. In the time interval between $t = 0$ and the instant when $f = 0$, the particle's velocity is
[TS EAMCET 2020]
(A) $\frac{1}{2} f_0 T^2$ (B) $f_0 T^2$
(C) $\frac{1}{2} f_0 T$ (D) $f_0 T$
- (39) A body initially at rest is moving with uniform acceleration a . Its velocity after n seconds is v . The displacement of the body in the last 2 s is
(A) $\frac{2v(n-1)}{n}$ (B) $\frac{v(n-1)}{n}$
(C) $\frac{v(n+1)}{n}$ (D) $\frac{2v(n+1)}{n}$
- (40) A balloon starts rising from the ground from rest with an upward acceleration of 2 m/s^2 . Just after 1 s, a stone is dropped from it. The time taken by the stone to strike the ground is nearly s.
(A) 0.3 (B) 0.7
(C) 1 (D) 1.4
- (41) A stone is dropped from a height h . It hits the ground with a certain momentum P . If the same stone is dropped from a height 100% more than the previous height, the momentum when it hits the ground will change by %.
[AIPMT 2012]
(A) 68 (B) 41
(C) 200 (D) 100
- (42) For a projectile, the maximum height and horizontal range are same. The angle of projection θ of the projectile is
[MHT CET 2024]
(A) $\tan^{-1}(\frac{1}{2})$ (B) $\tan^{-1}(2)$
(C) $\tan^{-1}(\frac{1}{4})$ (D) $\tan^{-1}(4)$
- (43) A car travels in a straight line along a road. Its distance x from a stop sign is given as a function of t by the equation $x(t) = \alpha t + \beta t^3$, where $\alpha = 2.0 \text{ m s}^{-1}$ and $\beta = 0.01 \text{ m s}^{-3}$.

Calculate the average velocity of the car in the time interval $t = 2.00 \text{ s}$ to $t = 4.00 \text{ s}$. (in m s^{-1})

[TS EAMCET 2021]

(A) 2.28 (B) 4.94

(C) 3.34 (D) 4.12

- (44) If a car at rest accelerates uniformly to a speed of 144 km/h in 20 s , then it covers a distance of m .

[AIPMT 1997]

(A) 20 (B) 400

(C) 1440 (D) 2880

- (45) A plane is flying horizontally at 98 m/s and releases an object which reaches the ground in 10 s . The angle made by the object with the horizontal while hitting the ground is $^\circ$.

(A) 55 (B) 45

(C) 60 (D) 75

- (46) Match the columns:

Column-I (R/H_{max})	Column-II (Angle of projection θ)
A.1	1.60°
B.4	2.30°
C. $4\sqrt{3}$	3.45°
D. $4/\sqrt{3}$	$4. \tan^{-1}(4) = 76^\circ$

(A) $A - 1, B - 2, C - 3, D - 4$ (B) $A - 4, B - 3, C - 2, D - 1$

(C) $A - 2, B - 1, C - 4, D - 3$ (D) $A - 3, B - 4, C - 1, D - 2$

- (47) The velocity of a body upon reaching the point from which it was projected upwards is:

[AIIMS 1999]

(A) $v = 0$ (B) $v = 2u$

(C) $v = 0.5u$ (D) $v = u$

- (48) A ball thrown by one player reaches the other in 2 s . The maximum height attained by the ball above the point of projection will be about m .

(A) 2.5 (B) 5

(C) 7.5 (D) 10

- (49) A projectile is projected with kinetic energy K . If it has the maximum possible horizontal range, then its kinetic energy at the highest point will be K .

(A) 0.25 (B) 0.5

(C) 0.75 (D) 1.0

- (50) A gun is aimed at a target in the line of its barrel. The target is released and allowed to fall under gravity at the same instant the gun is fired. The bullet will

(A) Pass above the target

(B) Pass below the target

(C) Hit the target

(D) Certainly miss the target



Devane physics class - 9545217210

Subject : Physics

Standard : 11

Total Mark : 200

Kinematics

(Answer Key)

Paper Set : 1

Date : 24-08-2026

Time : 0H:0M

Physics - Section A (MCQ)

1 - D	2 - D	3 - C	4 - D	5 - C	6 - C	7 - B	8 - A	9 - B	10 - A
11 - B	12 - B	13 - D	14 - A	15 - A	16 - D	17 - B	18 - C	19 - C	20 - C
21 - C	22 - B	23 - B	24 - B	25 - D	26 - D	27 - B	28 - B	29 - A	30 - D
31 - B	32 - C	33 - C	34 - C	35 - B	36 - C	37 - C	38 - C	39 - A	40 - B
41 - B	42 - D	43 - A	44 - B	45 - B	46 - B	47 - D	48 - B	49 - B	50 - C

Devane physics class



Subject : Physics

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Kinematics

(Solutions)

Paper Set : 1

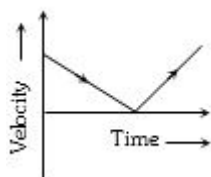
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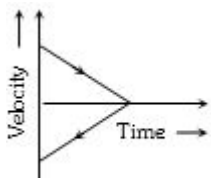
Physics - Section A (MCQ)

- (1) Which of the following represents the velocity-time graph for a ball thrown vertically upwards (neglecting air resistance)?

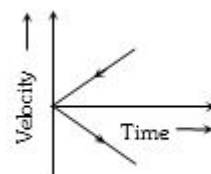
(A)



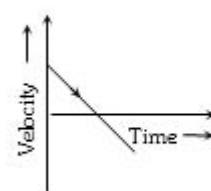
(B)



(C)



(D)



Solution:(Correct Answer:D)

(D) When a ball is thrown vertically upwards with an initial velocity u , it moves under the influence of constant gravitational acceleration g acting downwards.

Taking the upward direction as positive, the velocity at any time t is given by $v = u - gt$.

This is a linear equation of the form $y = mx + c$, where the slope is $-g$ (a constant negative value).

Initially, the velocity is positive and decreases linearly as the ball rises.

At the maximum height, the velocity becomes zero.

After reaching the maximum height, the ball starts moving downwards, so its velocity becomes negative and continues to decrease (become more negative) at a constant rate due to gravity.

Therefore, the velocity-time graph is a straight line with a

constant negative slope that passes through the time axis, representing the transition from upward motion (positive velocity) to downward motion (negative velocity). This corresponds to the graph shown in option D.

- (2) If the kinetic energy of a second ball at its maximum height is K , what will be the kinetic energy of the first ball at its maximum height? (Assuming the first ball is thrown vertically upwards).

(A) $4K$

(B) $3K$

(C) $2K$

(D) 0

Solution:(Correct Answer:D)

(D) The kinetic energy of a projectile at its maximum height is given by $KE = \frac{1}{2}m(u \cos \theta)^2$, where u is the initial velocity and θ is the angle of projection.

For a ball thrown vertically upwards, the angle of projection $\theta = 90^\circ$.

At the maximum height, the vertical component of velocity becomes zero.

Since $\cos 90^\circ = 0$, the kinetic energy at the maximum height for a ball thrown vertically is $KE = \frac{1}{2}m(u \cos 90^\circ)^2 = 0$.

Therefore, the kinetic energy of the first ball at its maximum height is 0.

- (3) The horizontal range is four times the maximum height attained by a projectile. The angle of projection is °

(A) 90

(B) 60

(C) 45

(D) 30

Solution:(Correct Answer:C)

(C) Given that the horizontal range R is four times the maximum height H , i.e., $R = 4H$.

The formula for horizontal range is $R = \frac{U^2 \sin(2\theta)}{g} = \frac{2U^2 \sin \theta \cos \theta}{g}$.

The formula for maximum height is $H = \frac{U^2 \sin^2 \theta}{2g}$.

Substituting these into the given condition: $\frac{2U^2 \sin \theta \cos \theta}{g} = 4 \times \frac{U^2 \sin^2 \theta}{2g}$.

Simplifying the equation: $2 \sin \theta \cos \theta = 2 \sin^2 \theta$.

Dividing both sides by $2 \sin \theta$ (assuming $\sin \theta \neq 0$): $\cos \theta = \sin \theta$.

Therefore, $\tan \theta = 1$, which implies $\theta = 45^\circ$.

- (4) If the velocity v of a particle moving along a straight line decreases linearly with its displacement s from 20 m/s to a value approaching zero at $s = 30 \text{ m}$, then the acceleration of the particle at $s = 15 \text{ m}$ is

(A) $2/3 \text{ m/s}^2$

(B) $-2/3 \text{ m/s}^2$

(C) $20/3 \text{ m/s}^2$

(D) $-20/3 \text{ m/s}^2$

Solution:(Correct Answer:D)

(D) The velocity v decreases linearly with displacement s .

The equation of a straight line is $v = ms + c$.

At $s = 0, v = 20 \text{ m/s}$, so $c = 20$.

At $s = 30 \text{ m}$, $v = 0$, so $0 = m(30) + 20$, which gives $m = -20/30 = -2/3$.

Thus, the velocity function is $v = -\frac{2}{3}s + 20$.

The acceleration a is given by $a = v \frac{dv}{ds}$.

First, find $\frac{dv}{ds} = \frac{d}{ds}(-\frac{2}{3}s + 20) = -\frac{2}{3}$.

At $s = 15 \text{ m}$, the velocity is $v = -\frac{2}{3}(15) + 20 = -10 + 20 = 10 \text{ m/s}$.

Therefore, the acceleration $a = (10)(-\frac{2}{3}) = -\frac{20}{3} \text{ m/s}^2$.

- (5) Statement (I): An object subjected to velocities $\vec{v}_1$ and $\vec{v}_2$ has a resultant velocity with magnitude $|\vec{v}| = |\vec{v}_1| + |\vec{v}_2|$.
Statement (II): The magnitude of displacement is either less than or equal to the path length of an object between two points.

Statement (III): The instantaneous acceleration is the limiting value of the average acceleration as the time interval approaches zero.

Which of the following is correct?

[TS EAMCET 2022]

- (A) Statements I, II, III are true
(B) Statements I, II are true, but Statement III is false
(C) Statements II, III are true, but Statement I is false
(D) Statements I, II, III are false

Solution:(Correct Answer:C)

(C) Statement (I) is incorrect because the resultant velocity magnitude is given by the vector addition formula: $|\vec{v}| = \sqrt{|\vec{v}_1|^2 + |\vec{v}_2|^2 + 2|\vec{v}_1||\vec{v}_2|\cos\theta}$. It is only equal to the sum of magnitudes if the vectors are in the same direction ($\theta = 0^\circ$).
Statement (II) is correct because displacement is the shortest distance between two points, while path length is the total distance covered. Thus, displacement $\leq$ distance.
Statement (III) is correct by definition. Instantaneous acceleration is defined as $\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t}$.

- (6) A projectile is thrown with velocity u at an angle of 15° and has a range R . What will be the range of the projectile if it is thrown with velocity $2u$ at an angle of 45° (in $\cdot R$)?

- (A) 12 (B) 3
(C) 8 (D) 4

Solution:(Correct Answer:C)

(C) The formula for the horizontal range R of a projectile is given by $R = \frac{u^2 \sin(2\theta)}{g}$.

From this, we can see that $R \propto u^2 \sin(2\theta)$.

Let the initial range be $R_1 = R$, initial velocity be $u_1 = u$, and initial angle be $\theta_1 = 15^\circ$.

Let the final range be R_2 , final velocity be $u_2 = 2u$, and final angle be $\theta_2 = 45^\circ$.

Taking the ratio:

$$\frac{R_2}{R_1} = \left(\frac{u_2}{u_1}\right)^2 \left(\frac{\sin(2\theta_2)}{\sin(2\theta_1)}\right)$$

Substituting the values:

$$\frac{R_2}{R} = \left(\frac{2u}{u}\right)^2 \left(\frac{\sin(2 \times 45^\circ)}{\sin(2 \times 15^\circ)}\right)$$

$$\frac{R_2}{R} = (2)^2 \left(\frac{\sin(90^\circ)}{\sin(30^\circ)}\right)$$

$$\frac{R_2}{R} = 4 \left(\frac{1}{0.5}\right) = 4 \times 2 = 8$$

Therefore, $R_2 = 8R$.

- (7) A person can throw a ball up to a maximum range of 100 m . How high above the ground can he throw the same ball (in $\cdot \text{m}$)?

[JEE MAIN 2022]

- (A) 25 (B) 50
(C) 100 (D) 200

Solution:(Correct Answer:B)

(B) The maximum horizontal range $R_{\max}$ of a projectile is given by the formula $R_{\max} = \frac{u^2}{g}$, where u is the initial velocity and g is the acceleration due to gravity.

Given $R_{\max} = 100 \text{ m}$, we have $\frac{u^2}{g} = 100 \text{ m}$.

The maximum height $H_{\max}$ reached by a projectile when thrown vertically upwards is given by $H_{\max} = \frac{u^2}{2g}$.

Substituting the value of $\frac{u^2}{g}$ into the height formula:

$$H_{\max} = \frac{1}{2} \times \left(\frac{u^2}{g}\right) = \frac{1}{2} \times 100 \text{ m} = 50 \text{ m}.$$

Therefore, the person can throw the ball to a maximum height of 50 m .

- (8) A projectile object is thrown in the upward direction making an angle of 60° with the horizontal with velocity of 140 m/s . Then, the time after which its velocity makes an angle 30° with the horizontal is (Take, $g = 10 \text{ m/s}^2$)

[TS EAMCET 2021]

- (A) $\frac{14}{\sqrt{3}} \text{ s}$ (B) $7\sqrt{3} \text{ s}$
(C) $14\sqrt{3} \text{ s}$ (D) $\frac{7}{\sqrt{3}} \text{ s}$

Solution:(Correct Answer:A)

(A) The horizontal component of velocity remains constant throughout the projectile motion. Let the initial velocity be $v_0 = 140 \text{ m/s}$ at angle $\theta_0 = 60^\circ$. Let the velocity at time t be v at angle $\theta = 30^\circ$.

Horizontal component: $v_x = v_0 \cos 60^\circ = v \cos 30^\circ$.

$$140 \times \frac{1}{2} = v \times \frac{\sqrt{3}}{2} \Rightarrow v = \frac{140}{\sqrt{3}} \text{ m/s}.$$

Vertical component at time t : $v_y = v_0 \sin 60^\circ - gt = v \sin 30^\circ$.

$$140 \times \frac{\sqrt{3}}{2} - 10t = \frac{140}{\sqrt{3}} \times \frac{1}{2}.$$

$$70\sqrt{3} - 10t = \frac{70}{\sqrt{3}}.$$

$$10t = 70\sqrt{3} - \frac{70}{\sqrt{3}} = 70 \left(\frac{3-1}{\sqrt{3}}\right) = \frac{140}{\sqrt{3}}.$$

$$t = \frac{14}{\sqrt{3}} \text{ s}.$$

- (9) The ratio of minimum kinetic energies of two projectiles of the same mass is $4 : 1$. The ratio of the maximum heights attained by them is also $4 : 1$. The ratio of their ranges would be: (in $\cdot 1$)

- (A) 16 (B) 4
(C) 8 (D) 2

Solution:(Correct Answer:B)

(B) The minimum kinetic energy of a projectile is at the highest point, where the vertical component of velocity is zero. Thus, $K_{\min} = \frac{1}{2} m u_x^2$.

Given the ratio of minimum kinetic energies is $\frac{K_{\min 1}}{K_{\min 2}} =$

$$\frac{\frac{1}{2} m u_{x1}^2}{\frac{1}{2} m u_{x2}^2} = \frac{4}{1}, \text{ which implies } \frac{u_{x1}}{u_{x2}} = \frac{2}{1}.$$

The maximum height is given by $H = \frac{u_y^2}{2g}$.

Given the ratio of maximum heights is $\frac{H_1}{H_2} = \frac{u_{y1}^2}{u_{y2}^2} = \frac{4}{1}$, which implies $\frac{u_{y1}}{u_{y2}} = \frac{2}{1}$.

The range of a projectile is $R = \frac{2u_x u_y}{g}$.

Therefore, the ratio of the ranges is $\frac{R_1}{R_2} = \frac{u_{x1} u_{y1}}{u_{x2} u_{y2}} = \left(\frac{u_{x1}}{u_{x2}}\right) \times \left(\frac{u_{y1}}{u_{y2}}\right) = \frac{2}{1} \times \frac{2}{1} = \frac{4}{1}$.

- (10) The maximum horizontal range of a projectile is 400 m . What is the maximum height attained by it (in $\cdot \text{m}$)?

- (A) 100 (B) 200
(C) 400 (D) 800

Solution:(Correct Answer:A)

(A) The maximum horizontal range is given by $R_{\max} = \frac{u^2}{g} = 400 \text{ m}$.

The maximum height attained by a projectile is given by

$$H = \frac{u^2 \sin^2 \theta}{2g}.$$

For maximum horizontal range, the angle of projection is $\theta = 45^\circ$.

Substituting $\theta = 45^\circ$ into the height formula:

$$H = \frac{u^2 \sin^2 45^\circ}{2g} = \frac{u^2 (1/\sqrt{2})^2}{2g} = \frac{u^2}{2g \times 2} = \frac{u^2}{4g}.$$

Since $R_{\max} = \frac{u^2}{g} = 400 \text{ m}$, we have:

$$H = \frac{1}{4} \times R_{\max} = \frac{1}{4} \times 400 \text{ m} = 100 \text{ m}.$$

- (11) What is the path followed by a moving body, on which a constant force acts in a direction other than the initial velocity (i.e., excluding parallel and antiparallel directions)?

- (A) Straight line (B) Parabolic
(C) Circular (D) Elliptical

Solution:(Correct Answer:B)

(B) When a constant force acts on a moving body in a direction that is not parallel or antiparallel to its initial velocity, the body experiences a constant acceleration in that direction.

Let the initial velocity be $\vec{u}$ and the constant acceleration be $\vec{a}$.

The position of the body at any time t is given by $\vec{r}(t) = \vec{u}t + \frac{1}{2}\vec{a}t^2$.

This equation represents a quadratic relationship between displacement and time, which is the characteristic equation of a parabola.

Therefore, the path followed by the body is parabolic.

- (12) A particle moves along the x -axis as $x = 4(t-2) + a(t-2)^2$. Which of the following statements is true?

- (A) The initial velocity of the particle is 4.
(B) The acceleration of the particle is $2a$.
(C) The particle is at the origin at $t = 0$.
(D) None of the above.

Solution:(Correct Answer:B)

(B) Given the position equation: $x = 4(t-2) + a(t-2)^2$.

To find the velocity v , we differentiate x with respect to time t :

$$v = \frac{dx}{dt} = \frac{d}{dt}[4(t-2) + a(t-2)^2] = 4 + 2a(t-2).$$

To find the acceleration a_{acc} , we differentiate the velocity v with respect to time t :

$$a_{\text{acc}} = \frac{dv}{dt} = \frac{d}{dt}[4 + 2a(t-2)] = 2a.$$

Checking the options:

(a) Initial velocity at $t = 0$ is $v = 4 + 2a(0-2) = 4 - 4a$.

This is not 4 unless $a = 0$.

(b) The acceleration is $2a$, which matches our derivation.

(c) At $t = 0$, $x = 4(0-2) + a(0-2)^2 = -8 + 4a$. This is not 0 unless $a = 2$.

Therefore, the correct statement is that the acceleration of the particle is $2a$.

- (13) Two paper screens A and B are separated by a distance of 100 m . A bullet penetrates A and B at points P and Q respectively, where Q is 10 cm below P . If the bullet is travelling horizontally at the time of hitting A , the velocity of the bullet at A is nearly m/s .

- (A) 100 (B) 200
(C) 600 (D) 700

Solution:(Correct Answer:D)

(D) Given:

Horizontal distance $x = 100 \text{ m}$

Vertical displacement $y = 10 \text{ cm} = 0.1 \text{ m}$

Initial vertical velocity $u_y = 0$

Acceleration due to gravity $g = 10 \text{ m/s}^2$

For horizontal projectile motion, the vertical displacement y is given by:

$$y = \frac{1}{2}gt^2$$

$$0.1 = \frac{1}{2} \times 10 \times t^2$$

$$0.1 = 5t^2$$

$$t^2 = 0.02$$

$$t = \sqrt{0.02} \text{ s}$$

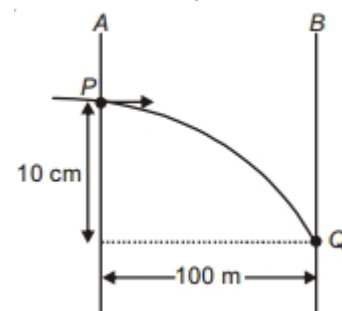
The horizontal distance x is covered with constant horizontal velocity u :

$$x = u \times t$$

$$100 = u \times \sqrt{0.02}$$

$$u = \frac{100}{\sqrt{0.02}} = \frac{100}{\sqrt{2 \times 10^{-2}}} = \frac{100}{0.1414} \approx 707 \text{ m/s}$$

Thus, the velocity of the bullet at A is nearly 700 m/s .



- (14) A particle is projected horizontally from a tower with velocity 10 m/s . Taking $g = 10 \text{ m/s}^2$, match the following two columns at time $t = 1 \text{ s}$.

Column I	Column II
(A) Horizontal component of velocity	(p) 5 SI unit
(B) Vertical component of velocity	(q) 10 SI unit
(C) Horizontal displacement	(r) 15 SI unit
(D) Vertical displacement	(s) 20 SI unit

- (A) $(A \rightarrow q, B \rightarrow q, C \rightarrow q, D \rightarrow p)$ (B) $(A \rightarrow q, B \rightarrow r, C \rightarrow q, D \rightarrow p)$
(C) $(A \rightarrow q, B \rightarrow s, C \rightarrow q, D \rightarrow p)$ (D) $(A \rightarrow s, B \rightarrow q, C \rightarrow q, D \rightarrow p)$

Solution:(Correct Answer:A)

(A) Given:

Initial horizontal velocity $u_x = 10 \text{ m/s}$.

Initial vertical velocity $u_y = 0 \text{ m/s}$.

Acceleration due to gravity $g = 10 \text{ m/s}^2$.

Time $t = 1 \text{ s}$.

(A) Horizontal component of velocity (v_x): In projectile motion, horizontal velocity remains constant. So, $v_x = u_x = 10 \text{ m/s}$. Matches with (q).

(B) Vertical component of velocity (v_y): Using $v_y = u_y + g_y t$, where $g_y = g = 10 \text{ m/s}^2$. So, $v_y = 0 + 10(1) = 10 \text{ m/s}$. Matches with (q).

(C) Horizontal displacement (x): $x = u_x t = 10(1) = 10 \text{ m}$. Matches with (q).

(D) Vertical displacement (y): $y = u_y t + \frac{1}{2}gt^2 = 0(1) +$

$$\frac{1}{2}(10)(1)^2 = 5 \text{ m. Matches with (p).}$$

Therefore, the correct matching is $(A \rightarrow q, B \rightarrow q, C \rightarrow q, D \rightarrow p)$.

- (15) When a ball is thrown with a velocity of 50 ms^{-1} at an angle 30° with the horizontal, it remains in the air for how many seconds? (Take $g = 10 \text{ ms}^{-2}$)

[AP EAMCET 2021]

- (A) 5 (B) 2.5
(C) 1.25 (D) 0.625

Solution:(Correct Answer:A)

(A) The time of flight T for a projectile is given by the formula $T = \frac{2u \sin \theta}{g}$.

Given:

Initial velocity $u = 50 \text{ ms}^{-1}$

Angle of projection $\theta = 30^\circ$

Acceleration due to gravity $g = 10 \text{ ms}^{-2}$

Substituting the values into the formula:

$$T = \frac{2 \times 50 \times \sin 30^\circ}{10}$$

$$\text{Since } \sin 30^\circ = 0.5,$$

$$T = \frac{100 \times 0.5}{10} = \frac{50}{10} = 5 \text{ s.}$$

Thus, the ball remains in the air for 5 s.

- (16) The position of an object moving along the x -axis is given by $x = a + bt^2$, where $a = 8.5 \text{ m}$, $b = 2.5 \text{ ms}^{-2}$, and t is measured in seconds. What is the average velocity between $t = 2.0 \text{ s}$ and $t = 4.0 \text{ s}$ (in ms^{-1})?

- (A) 20 (B) 10
(C) 25 (D) 15

Solution:(Correct Answer:D)

(D) The average velocity is defined as the change in position divided by the change in time: $v_{avg} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}$.

Given $x(t) = a + bt^2$, we calculate the positions at $t_1 = 2.0 \text{ s}$ and $t_2 = 4.0 \text{ s}$.

$$x(2.0) = a + b(2.0)^2 = a + 4b.$$

$$x(4.0) = a + b(4.0)^2 = a + 16b.$$

Now, substitute these into the average velocity formula:

$$v_{avg} = \frac{(a + 16b) - (a + 4b)}{4.0 - 2.0} = \frac{12b}{2.0} = 6b.$$

Given $b = 2.5 \text{ ms}^{-2}$, we have:

$$v_{avg} = 6 \times 2.5 = 15 \text{ ms}^{-1}.$$

- (17) A particle moves in the east direction with 15 m/s for 2 s , then moves northward with 5 m/s for 8 s . The average velocity of the particle is:

- (A) 5 m/s due $E - 37^\circ - N$
(B) 5 m/s due $N - 37^\circ - E$
(C) 7 m/s due $S - 37^\circ - W$
(D) 10 m/s due $N - 37^\circ - E$

Solution:(Correct Answer:B)

(B) The average velocity is defined as the total displacement divided by the total time taken.

First, calculate the displacement in the east direction:

$$\vec{d}_1 = 15 \text{ m/s} \times 2 \text{ s} \times \hat{i} = 30\hat{i} \text{ m}$$

Next, calculate the displacement in the north direction:

$$\vec{d}_2 = 5 \text{ m/s} \times 8 \text{ s} \times \hat{j} = 40\hat{j} \text{ m}$$

$$\text{Total displacement } \vec{d} = \vec{d}_1 + \vec{d}_2 = 30\hat{i} + 40\hat{j} \text{ m}$$

$$\text{Total time } t = 2 \text{ s} + 8 \text{ s} = 10 \text{ s}$$

$$\text{Average velocity } \vec{v}_{av} = \frac{\vec{d}}{t} = \frac{30\hat{i} + 40\hat{j}}{10} = 3\hat{i} + 4\hat{j} \text{ m/s}$$

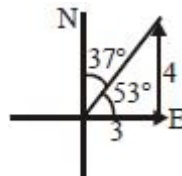
The magnitude of the average velocity is $|\vec{v}_{av}| =$

$$\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5 \text{ m/s.}$$

The direction is given by $\tan \theta = \frac{v_y}{v_x} = \frac{4}{3}$, which implies $\theta =$

53° north of east, or 37° east of north.

Thus, the average velocity is 5 m/s due $N - 37^\circ - E$.



- (18) What is the angle between the velocity and acceleration of a projectile at its maximum height (in $^\circ$)?

- (A) 0 (B) 45
(C) 90 (D) 180

Solution:(Correct Answer:C)

(C) At the maximum height of a projectile, the vertical component of velocity (v_y) becomes 0, while the horizontal component (v_x) remains constant ($u \cos \theta$).

Thus, the resultant velocity vector is purely horizontal.

The acceleration of a projectile is always the acceleration due to gravity (g), which acts vertically downwards.

Since the velocity is horizontal and the acceleration is vertical, the angle between them is 90° .

- (19) If a stone is released from a balloon rising with acceleration a at the instant when its velocity is v , then immediately after release, the acceleration and velocity of the stone are

- (A) a (upward), v (upward)
(B) g (upward), v (upward)
(C) g (downward), v (upward)
(D) $(g - a)$ (downward), v (upward)

Solution:(Correct Answer:C)

(C) 1. At the instant the stone is released, it possesses the same velocity as the balloon due to inertia. Therefore, the velocity of the stone immediately after release is v (upward).

2. Once the stone is released, it is no longer in contact with the balloon and is only under the influence of gravity.

3. The only force acting on the stone is its weight, which acts vertically downward.

4. According to Newton's second law, the acceleration of the stone is $F/m = mg/m = g$.

5. Thus, the acceleration of the stone is g (downward).

- (20) A ball is projected upwards from a height h above the surface of the earth with velocity v . The time at which the ball strikes the ground is

[AP EAMCET 2020]

- (A) $\frac{v}{g} \left[1 + \sqrt{\frac{2gh}{v^2}} \right]$ (B) $\frac{v}{g} \left[1 - \sqrt{1 + \frac{2h}{g}} \right]$
(C) $\frac{v}{g} \left[1 + \sqrt{1 + \frac{2gh}{v^2}} \right]$ (D) $\frac{v}{g} \left[1 + \sqrt{v^2 + \frac{2g}{v^2}} \right]$

Solution:(Correct Answer:C)

(C) Let the upward direction be positive and the downward direction be negative. The initial position is $y_0 = h$ and the final position is $y = 0$.

The initial velocity is $u = v$ and the acceleration is $a = -g$.

Using the equation of motion $y = y_0 + ut + \frac{1}{2}at^2$, we get:

$$0 = h + vt - \frac{1}{2}gt^2$$

Rearranging the terms, we get the quadratic equation:

$$\frac{1}{2}gt^2 - vt - h = 0$$

Multiplying by 2, we get $gt^2 - 2vt - 2h = 0$.

Using the quadratic formula $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a =$

$$g, b = -2v, \text{ and } c = -2h:$$

$$t = \frac{2v \pm \sqrt{(-2v)^2 - 4(g)(-2h)}}{2g}$$

$$t = \frac{2v \pm \sqrt{4v^2 + 8gh}}{2g}$$

$$t = \frac{2v \pm 2\sqrt{v^2 + 2gh}}{2g}$$

$$t = \frac{v \pm \sqrt{v^2 + 2gh}}{g}$$

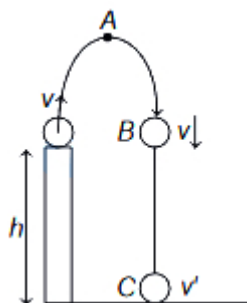
Since time must be positive, we take the positive root:

$$t = \frac{v + \sqrt{v^2 + 2gh}}{g}$$

$$t = \frac{v}{g} + \frac{\sqrt{v^2(1 + \frac{2gh}{v^2})}}{g}$$

$$t = \frac{v}{g} + \frac{v}{g} \sqrt{1 + \frac{2gh}{v^2}}$$

$$t = \frac{v}{g} \left[1 + \sqrt{1 + \frac{2gh}{v^2}} \right]$$



- (21) A particle covers half of its total distance with speed v_1 and the rest half distance with speed v_2 . Its average speed during the complete journey is [AIPMT 2011]

(A) $\frac{v_1 + v_2}{2}$

(B) $\frac{v_1 v_2}{v_1 + v_2}$

(C) $\frac{2v_1 v_2}{v_1 + v_2}$

(D) $\frac{v_1 + v_2}{3}$

Solution:(Correct Answer:C)

(C) Let the total distance be d . The particle covers the first half distance ($d/2$) with speed v_1 and the second half distance ($d/2$) with speed v_2 .

Time taken for the first half, $t_1 = \frac{d/2}{v_1} = \frac{d}{2v_1}$.

Time taken for the second half, $t_2 = \frac{d/2}{v_2} = \frac{d}{2v_2}$.

Average speed is defined as the total distance divided by the total time taken:

$$v_{av} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{d}{t_1 + t_2}$$

Substituting the values of t_1 and t_2 :

$$v_{av} = \frac{d}{\frac{d}{2v_1} + \frac{d}{2v_2}} = \frac{d}{\frac{d}{2} \left(\frac{1}{v_1} + \frac{1}{v_2} \right)} = \frac{1}{\frac{1}{2} \left(\frac{1}{v_1} + \frac{1}{v_2} \right)}$$

$$v_{av} = \frac{2v_1 v_2}{v_1 + v_2}$$

- (22) A particle of mass 1 kg is moving along the x -axis, where its velocity depends on the x -coordinate as $v = 2\sqrt{x} \text{ m/s}$. Find the net force acting on the particle 2 s after the start of motion.

(A) 1

(B) 2

(C) 4

(D) 1.5

Solution:(Correct Answer:B)

(B) Given, $v = 2\sqrt{x}$.

Acceleration $a = v \frac{dv}{dx}$.

First, find $\frac{dv}{dx}$:

$$\frac{dv}{dx} = \frac{d}{dx} (2x^{1/2}) = 2 \cdot \frac{1}{2} x^{-1/2} = \frac{1}{\sqrt{x}}$$

Now, calculate acceleration:

$$a = (2\sqrt{x}) \cdot \left(\frac{1}{\sqrt{x}} \right) = 2 \text{ m/s}^2$$

Since the acceleration is constant and independent of time and position, the net force F acting on the particle is given by Newton's second law:

$$F = m \cdot a = 1 \text{ kg} \cdot 2 \text{ m/s}^2 = 2 \text{ N}$$

Thus, the net force at any instant, including at $t = 2 \text{ s}$, is 2 N .

- (23) A body is falling freely from a point A at a certain height from the ground and passes through points B , C , and D

(vertically as shown below) such that $BC = CD$. The time taken by the particle to move from B to C is 2 s and from C to D is 1 s . The time taken to move from A to B in seconds is [TS EAMCET 2016]



(A) 0.6

(B) 0.5

(C) 0.2

(D) 0.4

Solution:(Correct Answer:B)

(B) Let the velocity of the particle at point B be v .

Since the body is falling freely, the acceleration is g .

For the interval BC , the time taken is $t_1 = 2 \text{ s}$. Using the equation of motion $s = ut + \frac{1}{2}at^2$:

$$BC = v(2) + \frac{1}{2}g(2)^2 = 2v + 2g \quad (i)$$

For the interval BD , the total time is $t_2 = 2 + 1 = 3 \text{ s}$.

$$BD = BC + CD = 2BC \quad (\text{since } BC = CD).$$

$$BD = v(3) + \frac{1}{2}g(3)^2 = 3v + 4.5g$$

Since $BD = 2BC$, we have:

$$2(2v + 2g) = 3v + 4.5g$$

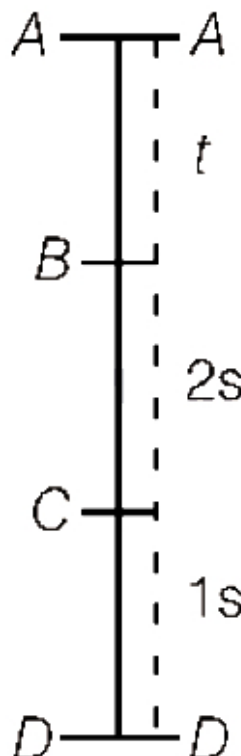
$$4v + 4g = 3v + 4.5g$$

$$v = 0.5g$$

Now, for the interval AB , let the time taken be t . Since the body falls freely from A , its initial velocity $u = 0$.

$$v = u + at \Rightarrow 0.5g = 0 + gt$$

$$t = 0.5 \text{ s}$$



- (24) A train has a speed of 60 km/h for the first one hour and 40 km/h for the next half hour. Its average speed in km/h is

(A) 50 (B) 53.33
(C) 48 (D) 70

Solution:(Correct Answer:B)

(B) The distance travelled by the train in the first 1 h is $d_1 = 60 \text{ km/h} \times 1 \text{ h} = 60 \text{ km}$.

The distance travelled by the train in the next 0.5 h is $d_2 = 40 \text{ km/h} \times 0.5 \text{ h} = 20 \text{ km}$.

The total distance travelled is $D = d_1 + d_2 = 60 \text{ km} + 20 \text{ km} = 80 \text{ km}$.

The total time taken is $T = 1 \text{ h} + 0.5 \text{ h} = 1.5 \text{ h} = 3/2 \text{ h}$.

The average speed is given by Average Speed =

$$\frac{\text{Total Distance}}{\text{Total Time}} = \frac{80}{1.5} = \frac{80}{3/2} = \frac{160}{3} \approx 53.33 \text{ km/h}.$$

- (25) When the velocity of a body is variable, then:

(A) Its speed may be constant
(B) Its acceleration may be constant
(C) Its average acceleration may be constant
(D) All of these

Solution:(Correct Answer:D)

(D) The correct answer is (d).

Velocity is a vector quantity, meaning it has both magnitude (speed) and direction. If the velocity is variable, it means either the magnitude is changing, the direction is changing, or both are changing.

(i) If a body moves in a uniform circular motion, its speed (magnitude of velocity) remains constant, but its direction changes continuously. Thus, the velocity is variable while the speed is constant.

(ii) In the case of uniform circular motion, the acceleration (centripetal acceleration) has a constant magnitude, even though its direction changes. Alternatively, for a body under constant acceleration (like gravity), the velocity changes linearly, which is also a variable velocity.

(iii) Average acceleration is defined as $a_{av} = \frac{v_2 - v_1}{t_2 - t_1}$. This value can be constant depending on the nature of the velocity change over the given time interval. Since all three conditions are possible, the correct option is (d).

- (26) A piece of marble is projected from the Earth's surface with a velocity of $19.6\sqrt{2} \text{ m/s}$ at an angle of 45° . 2 s later, its velocity makes an angle α with the horizontal, where α is (in $^\circ$)

(A) 45 (B) 30
(C) 60 (D) 0

Solution:(Correct Answer:D)

(D) The initial velocity is $u = 19.6\sqrt{2} \text{ m/s}$ at an angle $\theta = 45^\circ$.

Horizontal component: $u_x = u \cos(45^\circ) = 19.6\sqrt{2} \times \frac{1}{\sqrt{2}} = 19.6 \text{ m/s}$.

Vertical component: $u_y = u \sin(45^\circ) = 19.6\sqrt{2} \times \frac{1}{\sqrt{2}} = 19.6 \text{ m/s}$.

After time $t = 2 \text{ s}$, the horizontal velocity remains constant: $v_x = u_x = 19.6 \text{ m/s}$.

The vertical velocity is given by $v_y = u_y - gt = 19.6 - (9.8 \times 2) = 19.6 - 19.6 = 0 \text{ m/s}$.

Since the vertical component of velocity is 0 m/s , the velocity vector is purely horizontal.

Therefore, the angle α with the horizontal is 0° .

- (27) The ratio of times taken by a freely falling body to travel the first 5 m , second 5 m , and third 5 m distances is:

[TS EAMCET 2025]

(A) $1 : \sqrt{2} : \sqrt{3}$ (B) $1 : \sqrt{2} - 1 : \sqrt{3} - \sqrt{2}$
(C) $1 : \sqrt{3} : \sqrt{5}$ (D) $1 : 2 : 3$

Solution:(Correct Answer:B)

(B) For a body falling freely from rest, the distance s covered in time t is given by $s = \frac{1}{2}gt^2$.

Thus, the time taken to cover a distance s is $t = \sqrt{\frac{2s}{g}}$.

Let t_1 be the time to travel the first 5 m ($s_1 = 5 \text{ m}$). Then

$$t_1 = \sqrt{\frac{2(5)}{g}} = \sqrt{\frac{10}{g}}.$$

Let t_2 be the time to travel the first 10 m ($s_2 = 10 \text{ m}$). Then

$$t_2 = \sqrt{\frac{2(10)}{g}} = \sqrt{\frac{20}{g}}.$$

Let t_3 be the time to travel the first 15 m ($s_3 = 15 \text{ m}$). Then

$$t_3 = \sqrt{\frac{2(15)}{g}} = \sqrt{\frac{30}{g}}.$$

The time taken to travel the first 5 m is $T_1 = t_1 = \sqrt{\frac{10}{g}}$.

The time taken to travel the second 5 m is $T_2 = t_2 - t_1 = \sqrt{\frac{20}{g}} - \sqrt{\frac{10}{g}} = \sqrt{\frac{10}{g}}(\sqrt{2} - 1)$.

The time taken to travel the third 5 m is $T_3 = t_3 - t_2 = \sqrt{\frac{30}{g}} - \sqrt{\frac{20}{g}} = \sqrt{\frac{10}{g}}(\sqrt{3} - \sqrt{2})$.

The ratio $T_1 : T_2 : T_3$ is $\sqrt{\frac{10}{g}} : \sqrt{\frac{10}{g}}(\sqrt{2} - 1) : \sqrt{\frac{10}{g}}(\sqrt{3} - \sqrt{2})$.

Simplifying, we get $1 : (\sqrt{2} - 1) : (\sqrt{3} - \sqrt{2})$.

- (28) An object is projected such that it attains maximum range. Another body is projected to reach maximum height. If both objects reach the same maximum height, what is the ratio of their initial velocities?

[AP EAMCET 2024]

(A) $2 : 1$ (B) $\sqrt{2} : 1$
(C) $1 : \sqrt{2}$ (D) $1 : 2$

Solution:(Correct Answer:B)

(B) For the first object, the range is maximum when the angle of projection $\theta = 45^\circ$.

The maximum height attained by the first object is $H_1 = \frac{u_1^2 \sin^2 45^\circ}{2g} = \frac{u_1^2 (1/\sqrt{2})^2}{2g} = \frac{u_1^2}{4g}$.

For the second object, the maximum height is attained when it is projected vertically, i.e., $\theta = 90^\circ$.

The maximum height attained by the second object is $H_2 = \frac{u_2^2 \sin^2 90^\circ}{2g} = \frac{u_2^2}{2g}$.

Given that both objects reach the same maximum height, $H_1 = H_2$.

Therefore, $\frac{u_1^2}{4g} = \frac{u_2^2}{2g}$.

This simplifies to $\frac{u_1^2}{u_2^2} = \frac{4g}{2g} = 2$.

Taking the square root, we get $\frac{u_1}{u_2} = \sqrt{2} : 1$.

- (29) A ball is dropped downwards. After 1 second , another ball is dropped downwards from the same point. What is the distance between them after 3 seconds ?

(A) 25 (B) 20
(C) 50 (D) 9.8

Solution:(Correct Answer:A)

(A) Let the first ball be B_1 and the second ball be B_2 .

B_1 is dropped at $t = 0$. The distance covered by B_1 in 3 seconds is $s_1 = \frac{1}{2}g(3)^2 = 4.5g$.

B_2 is dropped at $t = 1 \text{ second}$. Therefore, at $t = 3 \text{ seconds}$, B_2 has been falling for 2 seconds . The distance covered by

$$B_2 \text{ is } s_2 = \frac{1}{2}g(2)^2 = 2g.$$

Taking $g = 10 \text{ m/s}^2$, we get $s_1 = 45 \text{ m}$ and $s_2 = 20 \text{ m}$.

The distance between them is $s_1 - s_2 = 45 - 20 = 25 \text{ m}$.

- (30) An aeroplane moving horizontally with a speed of 720 km/h drops a food packet while flying at a height of 396.9 m . The time taken by the food packet to reach the ground and its horizontal range are (Take $g = 9.8 \text{ m/s}^2$):

- (A) 3 s and 2000 m (B) 5 s and 500 m
(C) 8 s and 1500 m (D) 9 s and 1800 m

Solution:(Correct Answer:D)

(D) Given: Horizontal velocity $u = 720 \text{ km/h} = 720 \times \frac{5}{18} \text{ m/s}$
 $s = 200 \text{ m/s}$.

Height $h = 396.9 \text{ m}$.

Acceleration due to gravity $g = 9.8 \text{ m/s}^2$.

Time taken to reach the ground (t):

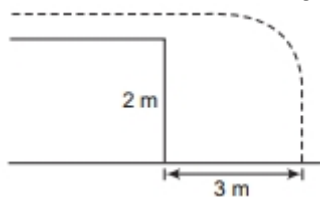
$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 396.9}{9.8}} = \sqrt{\frac{793.8}{9.8}} = \sqrt{81} = 9 \text{ s}.$$

Horizontal range (R):

$$R = u \times t = 200 \text{ m/s} \times 9 \text{ s} = 1800 \text{ m}.$$

Therefore, the time taken is 9 s and the horizontal range is 1800 m .

- (31) Water is flowing from a horizontal pipe fixed at a height of 2 m from the ground. If it falls at a horizontal distance of 3 m as shown in the figure, the speed of water when it leaves the pipe is ms^{-1} (take $g = 9.8 \text{ ms}^{-2}$).



- (A) 2.4 (B) 4.7
(C) 7.4 (D) 6.2

Solution:(Correct Answer:B)

(B) The water follows a projectile motion path after leaving the horizontal pipe.

Given:

Height of the pipe, $h = 2 \text{ m}$

Horizontal range, $R = 3 \text{ m}$

Acceleration due to gravity, $g = 9.8 \text{ ms}^{-2}$

Step 1: Calculate the time taken (t) for the water to reach the ground.

Using the equation of motion for vertical displacement: $h = \frac{1}{2}gt^2$

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 2}{9.8}} = \sqrt{\frac{4}{9.8}} \approx 0.6389 \text{ s} \approx 0.64 \text{ s}$$

Step 2: Calculate the horizontal speed (v) of the water.

Since the horizontal motion is uniform, $R = v \times t$

$$v = \frac{R}{t} = \frac{3}{0.6389} \approx 4.695 \text{ ms}^{-1} \approx 4.7 \text{ ms}^{-1}$$

Therefore, the speed of water when it leaves the pipe is 4.7 ms^{-1} .

- (32) The angle of projection of a projectile for which its initial kinetic energy becomes half at its maximum height is (in $^\circ$) [AP EAMCET 2017]

- (A) 90 (B) 60
(C) 45 (D) 30

Solution:(Correct Answer:C)

(C) Let the initial velocity of the projectile be u and the angle of projection be θ . The initial kinetic energy is $K_i = \frac{1}{2}mu^2$.

At the maximum height, the vertical component of velocity

becomes zero, and the velocity of the projectile is only the horizontal component, $v_x = u \cos \theta$.

The kinetic energy at the maximum height is $K_f =$

$$\frac{1}{2}m(u \cos \theta)^2 = \frac{1}{2}mu^2 \cos^2 \theta.$$

According to the problem, $K_f = \frac{1}{2}K_i$.

Substituting the expressions, we get $\frac{1}{2}mu^2 \cos^2 \theta = \frac{1}{2}(\frac{1}{2}mu^2)$.

This simplifies to $\cos^2 \theta = \frac{1}{2}$, which means $\cos \theta = \frac{1}{\sqrt{2}}$.

Therefore, $\theta = 45^\circ$.

- (33) The time of flight of an object projected with speed 20 m/s at an angle 30° with the horizontal is s .

- (A) 1 (B) 4
(C) 2 (D) 6

Solution:(Correct Answer:C)

(C) The formula for the time of flight T of a projectile is given by $T = \frac{2v \sin \theta}{g}$.

Given: Initial speed $v = 20 \text{ m/s}$, angle of projection $\theta = 30^\circ$, and acceleration due to gravity $g = 10 \text{ m/s}^2$.

Substituting the values into the formula:

$$T = \frac{2 \times 20 \times \sin 30^\circ}{10}$$

Since $\sin 30^\circ = 0.5$ or $\frac{1}{2}$:

$$T = \frac{2 \times 20 \times 0.5}{10} = \frac{20}{10} = 2 \text{ s}.$$

Therefore, the correct option is C.

- (34) A particle initially at rest is moving along a straight line with an acceleration of 2 m/s^2 . At a time of 3 s after the beginning of motion, the direction of acceleration is reversed. The time from the beginning of the motion in which the particle returns to its initial position is [TS EAMCET 2025]

- (A) $(3 + \sqrt{3}) \text{ s}$ (B) $(2 + \sqrt{2}) \text{ s}$
(C) $3(1 + \sqrt{2}) \text{ s}$ (D) $2(3 + \sqrt{3}) \text{ s}$

Solution:(Correct Answer:C)

(C) 1. For the first 3 s , the particle starts from rest ($u = 0$) with acceleration $a_1 = 2 \text{ m/s}^2$.

2. Velocity at $t = 3 \text{ s}$ is $v = u + a_1t = 0 + 2(3) = 6 \text{ m/s}$.

3. Displacement at $t = 3 \text{ s}$ is $s_1 = ut + 0.5a_1t^2 = 0 + 0.5(2)(3^2) = 9 \text{ m}$.

4. After $t = 3 \text{ s}$, the acceleration reverses, so $a_2 = -2 \text{ m/s}^2$. Let the additional time be t' .

5. The position at time t' is $s = s_1 + vt' + 0.5a_2(t')^2$.

6. For the particle to return to its initial position, $s = 0$.

$$0 = 9 + 6t' - 0.5(2)(t')^2 \Rightarrow 0 = 9 + 6t' - (t')^2 \Rightarrow (t')^2 - 6t' - 9 = 0.$$

8. Using the quadratic formula $t' = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} =$

$$\frac{6 \pm \sqrt{36 - 4(1)(-9)}}{2} = \frac{6 \pm \sqrt{72}}{2} = 3 \pm 3\sqrt{2}.$$

9. Since $t' > 0$, $t' = 3 + 3\sqrt{2} = 3(1 + \sqrt{2}) \text{ s}$.

10. Total time $T = 3 + t' = 3 + 3 + 3\sqrt{2} = 6 + 3\sqrt{2} = 3(2 + \sqrt{2}) \text{ s}$.

- (35) A boy walks to his school at a distance of 6 km with a constant speed of 2.5 km/h and walks back with a constant speed of 4 km/h . His average speed for the round trip, expressed in km/h , is:

- (A) $24/13$ (B) $40/13$
(C) 3 (D) $1/2$

Solution:(Correct Answer:B)

(B) The average speed for a round trip where the distance for both legs is the same is given by the formula: $v_{avg} =$

$$\frac{2v_1v_2}{v_1 + v_2}.$$

Here, $v_1 = 2.5 \text{ km/h}$ and $v_2 = 4 \text{ km/h}$.

Substituting the values into the formula:

$$v_{avg} = \frac{2 \times 2.5 \times 4}{2.5 + 4}$$

$$v_{avg} = \frac{20}{6.5}$$

$$v_{avg} = \frac{200}{65} = \frac{40}{13} \text{ km/h.}$$

Thus, the correct option is B.

- (36) With what velocity should a ball be projected vertically so that the distance covered by it in the 5th second is twice the distance it covers in its 6th second? (Take $g = 10 \text{ m/s}^2$)

(A) 58.8 (B) 49
(C) 65 (D) 19.6

Solution:(Correct Answer:C)

(C) The distance covered in the n^{th} second for an object projected vertically upwards is given by the formula: $h_{n^{\text{th}}} = u - \frac{g}{2}(2n - 1)$.

For the 5th second ($n = 5$): $h_{5^{\text{th}}} = u - \frac{10}{2}(2 \times 5 - 1) = u - 5(9) = u - 45$.

For the 6th second ($n = 6$): $h_{6^{\text{th}}} = u - \frac{10}{2}(2 \times 6 - 1) = u - 5(11) = u - 55$.

According to the problem, the distance covered in the 5th second is twice the distance covered in the 6th second:

$$h_{5^{\text{th}}} = 2 \times h_{6^{\text{th}}}.$$

Substituting the values:

$$u - 45 = 2(u - 55).$$

$$u - 45 = 2u - 110.$$

$$u = 110 - 45 = 65 \text{ m/s.}$$

- (37) A ball of mass 0.2 kg is thrown from a height of 1 m with an initial velocity of $\sqrt{10} \text{ m/s}$ at an angle of 45° with the horizontal. Assuming acceleration due to gravity $g = 10 \text{ m/s}^2$, the modulus of the momentum increment during the total time of motion in $\text{kg} \cdot \text{m/s}$ is:

[TS EAMCET 2018]

(A) $\frac{2+\sqrt{10}}{\sqrt{10}}$ (B) $\frac{1+\sqrt{10}}{\sqrt{5}}$
(C) $\frac{1+\sqrt{5}}{\sqrt{5}}$ (D) $\frac{\sqrt{5}-1}{\sqrt{5}}$

Solution:(Correct Answer:C)

(C) The force acting on the ball is gravity, which acts only in the vertical (y) direction. Therefore, there is no change in momentum in the horizontal (x) direction.

Initial vertical velocity: $u_y = u \sin 45^\circ = \sqrt{10} \cdot \frac{1}{\sqrt{2}} = \sqrt{5} \text{ m/s.}$

Initial vertical momentum: $p_{yi} = mu_y = 0.2 \cdot \sqrt{5} = \frac{\sqrt{5}}{5} = \frac{1}{\sqrt{5}} \text{ kg} \cdot \text{m/s}$ (upwards).

Using the equation of motion $v_y^2 = u_y^2 + 2gh$ (taking downward as positive for final velocity calculation, or using energy conservation):

$$v_y^2 = (\sqrt{5})^2 + 2(10)(1) = 5 + 20 = 25.$$

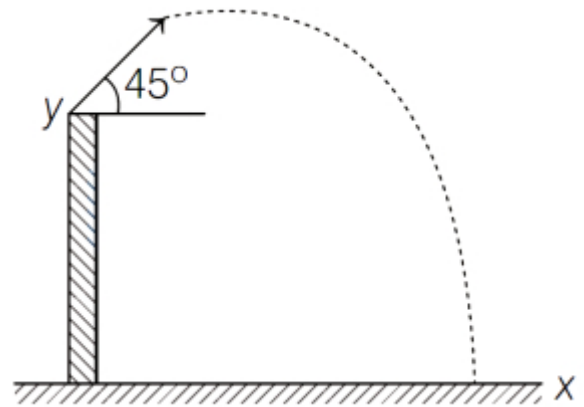
So, final vertical velocity $v_y = 5 \text{ m/s}$ (downwards).

Final vertical momentum: $p_{yf} = mv_y = 0.2 \cdot 5 = 1 \text{ kg} \cdot \text{m/s}$ (downwards).

Taking upward direction as positive, $p_{yi} = \frac{1}{\sqrt{5}}$ and $p_{yf} = -1$.

The change in momentum is $\Delta p = p_{yf} - p_{yi} = -1 - \frac{1}{\sqrt{5}} = -(1 + \frac{1}{\sqrt{5}})$.

The modulus of the momentum increment is $|\Delta p| = 1 + \frac{1}{\sqrt{5}} = \frac{\sqrt{5}+1}{\sqrt{5}}$.



- (38) A particle moving along the X -axis has acceleration f at time t given by $f = f_0(1 - \frac{t}{T})$, where f_0 and T are constants. The particle at $t = 0$ has zero velocity. In the time interval between $t = 0$ and the instant when $f = 0$, the particle's velocity is

[TS EAMCET 2020]

(A) $\frac{1}{2}f_0T^2$ (B) f_0T^2
(C) $\frac{1}{2}f_0T$ (D) f_0T

Solution:(Correct Answer:C)

(C) The acceleration of the particle is given by $f = f_0(1 - \frac{t}{T})$.

Since $f = \frac{dv}{dt}$, the velocity v is obtained by integrating the acceleration with respect to time:

$$v = \int f dt = \int f_0(1 - \frac{t}{T}) dt = f_0(t - \frac{t^2}{2T}) + C.$$

Given that at $t = 0, v = 0$, we find the constant of integration $C = 0$.

Thus, the velocity at any time t is $v = f_0(t - \frac{t^2}{2T})$.

The acceleration becomes zero when $f = 0$, which implies $1 - \frac{t}{T} = 0$, so $t = T$.

Substituting $t = T$ into the velocity equation:

$$v = f_0(T - \frac{T^2}{2T}) = f_0(T - \frac{T}{2}) = \frac{f_0T}{2}.$$

- (39) A body initially at rest is moving with uniform acceleration a . Its velocity after n seconds is v . The displacement of the body in the last 2 s is

(A) $\frac{2v(n-1)}{n}$ (B) $\frac{v(n-1)}{n}$
(C) $\frac{v(n+1)}{n}$ (D) $\frac{2v(n+1)}{n}$

Solution:(Correct Answer:A)

(A) Given that the body starts from rest, initial velocity $u = 0$. After n seconds, the velocity is v .

Using the first equation of motion, $v = u + an$, we get $v = 0 + an$, which implies $a = \frac{v}{n}$.

The displacement in the last 2 s is the difference between the total displacement in n seconds and the displacement in $(n - 2)$ seconds.

$$S_{\text{last } 2s} = S_n - S_{n-2} = \frac{1}{2}an^2 - \frac{1}{2}a(n-2)^2.$$

$$S_{\text{last } 2s} = \frac{1}{2}a[n^2 - (n^2 - 4n + 4)] = \frac{1}{2}a[4n - 4] = 2a(n-1).$$

Substituting $a = \frac{v}{n}$, we get $S_{\text{last } 2s} = 2(\frac{v}{n})(n-1) = \frac{2v(n-1)}{n}$.

- (40) A balloon starts rising from the ground from rest with an upward acceleration of 2 m/s^2 . Just after 1 s, a stone is dropped from it. The time taken by the stone to strike the ground is nearly s.

(A) 0.3 (B) 0.7
(C) 1 (D) 1.4

Solution:(Correct Answer:B)

(B) 1. Initial motion of the balloon: The balloon starts from rest ($u = 0$) with acceleration ($a = 2 \text{ m/s}^2$). After $t = 1 \text{ s}$, the velocity of the balloon is $v = u + at = 0 + 2(1) = 2 \text{ m/s}$. The height reached by the balloon is $h = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2)(1)^2 = 1 \text{ m}$.

2. Motion of the stone: When the stone is dropped, it has an initial velocity equal to the balloon's velocity at that instant, which is $u_s = 2 \text{ m/s}$ (upwards). The stone is at a height of 1 m and is subject to gravity ($g = 9.8 \text{ m/s}^2$ downwards).

3. Using the equation of motion for the stone: $s = u_s t - \frac{1}{2}gt^2$. Here, $s = -1 \text{ m}$ (since it falls to the ground below the starting point).

$$-1 = 2t - \frac{1}{2}(9.8)t^2$$

$$-1 = 2t - 4.9t^2$$

$$4.9t^2 - 2t - 1 = 0$$

4. Solving the quadratic equation using the quadratic formula $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$t = \frac{2 \pm \sqrt{(-2)^2 - 4(4.9)(-1)}}{2(4.9)} = \frac{2 \pm \sqrt{4 + 19.6}}{9.8} = \frac{2 \pm \sqrt{23.6}}{9.8}$$

Taking the positive root: $t = \frac{2 + 4.858}{9.8} \approx \frac{6.858}{9.8} \approx 0.7 \text{ s}$.

- (41) A stone is dropped from a height h . It hits the ground with a certain momentum P . If the same stone is dropped from a height 100% more than the previous height, the momentum when it hits the ground will change by %.

[AIPMT 2012]

(A) 68

(B) 41

(C) 200

(D) 100

Solution:(Correct Answer:B)

(B) When a stone is dropped from a height h , its velocity v upon hitting the ground is given by $v = \sqrt{2gh}$.

Thus, the initial momentum is $P = m\sqrt{2gh} \dots (i)$.

If the height is increased by 100%, the new height $h' = h + 100\% \text{ of } h = h + h = 2h$.

The new momentum P' is $P' = m\sqrt{2g(2h)} = m\sqrt{2gh} \times \sqrt{2} = P\sqrt{2}$.

Given $\sqrt{2} \approx 1.414$.

The percentage change in momentum is $\frac{P' - P}{P} \times 100\% =$

$$\frac{P\sqrt{2} - P}{P} \times 100\% = (\sqrt{2} - 1) \times 100\%.$$

$$= (1.414 - 1) \times 100\% = 0.414 \times 100\% = 41.4\% \approx 41\%.$$

- (42) For a projectile, the maximum height and horizontal range are same. The angle of projection θ of the projectile is [MHT CET 2024]

(A) $\tan^{-1}(\frac{1}{2})$

(B) $\tan^{-1}(2)$

(C) $\tan^{-1}(\frac{1}{4})$

(D) $\tan^{-1}(4)$

Solution:(Correct Answer:D)

(D) Given that the horizontal range (R) is equal to the maximum height (H).

$$R = \frac{u^2 \sin(2\theta)}{g} = \frac{2u^2 \sin \theta \cos \theta}{g}$$

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Equating R and H :

$$\frac{2u^2 \sin \theta \cos \theta}{g} = \frac{u^2 \sin^2 \theta}{2g}$$

Dividing both sides by $\frac{u^2 \sin \theta}{g}$ (assuming $\sin \theta \neq 0$):

$$2 \cos \theta = \frac{\sin \theta}{2}$$

$$\frac{\sin \theta}{\cos \theta} = 4$$

$$\tan \theta = 4$$

$$\theta = \tan^{-1}(4)$$

- (43) A car travels in a straight line along a road. Its distance x from a stop sign is given as a function of t by the equation $x(t) = \alpha t + \beta t^3$, where $\alpha = 2.0 \text{ m s}^{-1}$ and $\beta = 0.01 \text{ m s}^{-3}$.

Calculate the average velocity of the car in the time interval $t = 2.00 \text{ s}$ to $t = 4.00 \text{ s}$. (in m s^{-1})

[TS EAMCET 2021]

(A) 2.28

(B) 4.94

(C) 3.34

(D) 4.12

Solution:(Correct Answer:A)

(A) The average velocity v_{avg} is defined as the total change in displacement divided by the total time interval: $v_{avg} =$

$$\frac{\Delta x}{\Delta t} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}.$$

Given $x(t) = \alpha t + \beta t^3$ with $\alpha = 2.0 \text{ m s}^{-1}$ and $\beta = 0.01 \text{ m s}^{-3}$.

At $t_1 = 2.00 \text{ s}$: $x(2) = 2.0(2) + 0.01(2)^3 = 4.0 + 0.01(8) = 4.08 \text{ m}$.

At $t_2 = 4.00 \text{ s}$: $x(4) = 2.0(4) + 0.01(4)^3 = 8.0 + 0.01(64) = 8.64 \text{ m}$.

Now, calculate the average velocity: $v_{avg} = \frac{8.64 - 4.08}{4.00 - 2.00} = \frac{4.56}{2} = 2.28 \text{ m s}^{-1}$.

- (44) If a car at rest accelerates uniformly to a speed of 144 km/h in 20 s , then it covers a distance of m .

[AIPMT 1997]

(A) 20

(B) 400

(C) 1440

(D) 2880

Solution:(Correct Answer:B)

(B) Given: Initial velocity $u = 0 \text{ m/s}$, final velocity $v = 144 \text{ km/h} = 144 \times \frac{5}{18} \text{ m/s} = 40 \text{ m/s}$, and time $t = 20 \text{ s}$.

Using the first equation of motion, $v = u + at$:

$$40 = 0 + a \times 20 \Rightarrow a = 2 \text{ m/s}^2.$$

Now, using the second equation of motion, $s = ut + \frac{1}{2}at^2$:

$$s = 0 \times 20 + \frac{1}{2} \times 2 \times (20)^2 = 400 \text{ m}.$$

Therefore, the distance covered is 400 m .

- (45) A plane is flying horizontally at 98 m/s and releases an object which reaches the ground in 10 s . The angle made by the object with the horizontal while hitting the ground is $^\circ$.

(A) 55

(B) 45

(C) 60

(D) 75

Solution:(Correct Answer:B)

(B) The horizontal component of the velocity of the object remains constant throughout the motion: $v_H = 98 \text{ m/s}$.

The vertical component of the velocity at time $t = 10 \text{ s}$ is given by $v_V = u_V + gt$. Since the object is released horizontally, $u_V = 0$.

$$v_V = 0 + (9.8 \text{ m/s}^2)(10 \text{ s}) = 98 \text{ m/s}.$$

The angle θ made by the object with the horizontal when hitting the ground is given by $\tan \theta = \frac{v_V}{v_H}$.

$$\tan \theta = \frac{98}{98} = 1.$$

$$\text{Therefore, } \theta = \tan^{-1}(1) = 45^\circ.$$

- (46) Match the columns:

Column-I (R/H_{max})	Column-II (Angle of projection θ)
A.1	1.60°
B.4	2.30°
C. $4\sqrt{3}$	3.45°
D. $4/\sqrt{3}$	$4. \tan^{-1}(4) = 76^\circ$

(A) $A - 1, B - 2, C - 3, D - 4$ (B) $A - 4, B - 3, C - 2, D - 1$

(C) $A - 2, B - 1, C - 4, D - 3$ (D) $A - 3, B - 4, C - 1, D - 2$

Solution:(Correct Answer:B)

(B) The range R and maximum height H of a projectile are given by:

$$R = \frac{u^2 \sin(2\theta)}{g} = \frac{2u^2 \sin \theta \cos \theta}{g}$$

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Taking the ratio R/H :

$$\frac{R}{H} = \frac{2u^2 \sin \theta \cos \theta}{g} \times \frac{2g}{u^2 \sin^2 \theta} = 4 \cot \theta$$

$$\text{Therefore, } \frac{R}{H} = 4 \cot \theta \text{ or } \tan \theta = \frac{4H}{R}.$$

$$\text{For A. } R/H = 1: 1 = 4 \cot \theta \Rightarrow \tan \theta = 4 \Rightarrow \theta = \tan^{-1}(4) = 76^\circ \text{ (Matches 4).}$$

$$\text{For B. } R/H = 4: 4 = 4 \cot \theta \Rightarrow \tan \theta = 1 \Rightarrow \theta = 45^\circ \text{ (Matches 3).}$$

$$\text{For C. } R/H = 4\sqrt{3}: 4\sqrt{3} = 4 \cot \theta \Rightarrow \tan \theta = 1/\sqrt{3} \Rightarrow \theta = 30^\circ \text{ (Matches 2).}$$

$$\text{For D. } R/H = 4/\sqrt{3}: 4/\sqrt{3} = 4 \cot \theta \Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ \text{ (Matches 1).}$$

Thus, the correct matching is $A - 4, B - 3, C - 2, D - 1$.

- (47) The velocity of a body upon reaching the point from which it was projected upwards is:

[AIIMS 1999]

(A) $v = 0$

(B) $v = 2u$

(C) $v = 0.5u$

(D) $v = u$

Solution:(Correct Answer:D)

(D) When a body is projected upwards with an initial velocity u , it undergoes constant deceleration due to gravity (g).

At the maximum height, its velocity becomes 0.

As it falls back down to the point of projection, it undergoes constant acceleration due to gravity (g) over the same displacement.

Using the kinematic equation $v^2 = u^2 + 2as$, where $a = -g$ for the upward journey and $a = g$ for the downward journey, the final velocity v at the starting point will have the same magnitude as the initial velocity u , but in the opposite direction.

Therefore, the speed of the body upon returning to the point of projection is $v = u$.

- (48) A ball thrown by one player reaches the other in 2 s. The maximum height attained by the ball above the point of projection will be about m.

(A) 2.5

(B) 5

(C) 7.5

(D) 10

Solution:(Correct Answer:B)

(B) The time of flight T for a projectile is given by $T = \frac{2u \sin \theta}{g}$.

Given $T = 2$ s, we have $2 = \frac{2u \sin \theta}{g}$, which simplifies to $u \sin \theta = g$.

The maximum height H attained by the projectile is given by $H = \frac{u^2 \sin^2 \theta}{2g}$.

Substituting $u \sin \theta = g$ into the equation for H , we get $H = \frac{(g)^2}{2g} = \frac{g}{2}$.

Taking $g = 10 \text{ m/s}^2$, we find $H = \frac{10}{2} = 5 \text{ m}$.

- (49) A projectile is projected with kinetic energy K . If it has the maximum possible horizontal range, then its kinetic energy at the highest point will be K .

(A) 0.25

(B) 0.5

(C) 0.75

(D) 1.0

Solution:(Correct Answer:B)

(B) The horizontal range R is given by $R = \frac{v^2 \sin(2\theta)}{g}$. For maximum horizontal range, the angle of projection θ must be 45° .

The initial kinetic energy is $K = \frac{1}{2}mv^2$.

At the highest point of the trajectory, the vertical component of velocity is zero, and the horizontal component is $v_x = v \cos(\theta)$.

Substituting $\theta = 45^\circ$, we get $v_x = v \cos(45^\circ) = \frac{v}{\sqrt{2}}$.

The kinetic energy at the highest point is $K' = \frac{1}{2}mv_x^2 = \frac{1}{2}m \left(\frac{v}{\sqrt{2}} \right)^2$.

$$K' = \frac{1}{2}m \left(\frac{v^2}{2} \right) = \frac{1}{2} \left(\frac{1}{2}mv^2 \right) = \frac{K}{2} = 0.5K.$$

- (50) A gun is aimed at a target in the line of its barrel. The target is released and allowed to fall under gravity at the same instant the gun is fired. The bullet will

(A) Pass above the target

(B) Pass below the target

(C) Hit the target

(D) Certainly miss the target

Solution:(Correct Answer:C)

(C) Let the initial velocity of the bullet be u and the angle of projection be θ . The target is at a distance d from the gun.

The time taken for the bullet to reach the horizontal distance d is $t = d/(u \cos \theta)$.

In this time t , the vertical displacement of the bullet due to gravity is $y_1 = (1/2)gt^2$.

The vertical position of the bullet relative to the line of sight (which is at angle θ) is $y_2 = (u \sin \theta)t - (1/2)gt^2$.

Since the target is in the line of the barrel, the initial height of the target is $h = d \tan \theta = (u \sin \theta)t$.

The actual position of the target at time t is $h' = h - (1/2)gt^2 = (u \sin \theta)t - (1/2)gt^2$.

Since the vertical position of the bullet y_2 is equal to the vertical position of the target h' , the bullet will hit the target.